

# **Data Preprocessing – Questions & Answers**

## **1. What do we need cleaning?**

Data cleaning is important because it helps us remove errors, unnecessary entries, and duplicate values. If the same data appears multiple times, it can affect the model and make it harder for the machine to understand the data correctly.

## **2. Why do we need transformation?**

Data transformation means converting raw data into a more useful and suitable format. This makes the data easier for the machine learning algorithm to understand and process.

## **3. What do we do when missing values exist?**

Missing values do not always mean that the data is useless; sometimes they can carry meaningful information. However, when we need to fill a missing value, we can use methods such as the median, especially when the data contains outliers.

## **4. Does anything need scaling?**

Yes, some machine learning algorithms are sensitive to the size or magnitude of features. Scaling puts the features on a similar range. Two common methods are normalization and standardization.

## **5. How do you split the data?**

We should not let the model practice on the same data that we use for the final exam. That is why we split the dataset into training, validation, and testing data. The model learns from the training data, is checked and tuned using validation data, and is finally evaluated on the unseen testing data.